

Selection of Time Headway in Connected and Autonomous Vehicle Platoons Under Noisy V2V Communication

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Abstract—In this paper, we investigate the selection of time headway to ensure robust string stability in connected and autonomous vehicle platoons in the presence of signal noise in Vehicle-to-Vehicle (V2V) communication. In particular, we consider the effect of noise in communicated vehicle acceleration from the predecessor vehicle to the follower vehicle on the selection of the time headway in predecessor-follower type vehicle platooning with a Constant Time Headway Policy (CTHP). Employing a CTHP based control law for each vehicle that utilizes onboard sensors for measurement of position and velocity of the predecessor vehicle and wireless communication network for obtaining the acceleration of the predecessor vehicle, we investigate how the implementable time headway is affected by communicated signal noise. We derive constraints on the CTHP controller gains for predecessor acceleration, velocity error and spacing error and a lower bound on the time headway which will ensure robust string stability of the platoon against signal noise. We perform comparative numerical simulations on an example to illustrate the main results.

Index Terms—Connected and autonomous vehicle platoons, cooperative adaptive cruise control (CACC), constant time headway policy (CTHP), V2V communication, signal-to-noise ratio (SNR), robust string stability.

I. INTRODUCTION

THE deployment of connected and autonomous vehicle platoons has the potential to benefit transportation systems in a profound and comprehensive way [2], [3]. Initially, under the Adaptive Cruise Control (ACC) paradigm facilitated by the use of onboard sensors (such as radars) for measuring the velocities of and distances from adjacent vehicles, the vehicle platoon can maintain a constant inter-vehicular spacing, referred to as Constant Spacing Policy (CSP) [4]. Recently, the benefits of employing advanced vehicular communication technologies and modern communication protocols have been expounded in great detail in the literature; for example, Dedicated Short Range Communication (DSRC) [5], Long Term Evolution (LTE) [6], 5G [7], and V2V communication [8].

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In addition to the use of onboard sensors, advanced vehicular wireless communications can lead to a higher-level of connectivity by incorporating the acceleration information of other vehicles which has the potential to significantly improve safety, increase mobility and throughput, and reduce fuel consumption under the paradigm of Cooperative Adaptive Cruise Control (CACC) with Constant Time Headway Policy (CTHP) [9].

In addition to internal stability for each vehicle, the connected and autonomous vehicle platoon should also exhibit robust string stability, i.e., robustness of the platoon to signal noise, parasitic actuation lag, and external disturbances. In recent decades, there has been extensive research devoted to connected and autonomous vehicle platoons on a wide range of topics including controller design [10], [11], [12], [13], [14], [15], communication mechanisms [16], [17], [18], experimental validation in realistic environments [19], [20], mixed human and autonomous traffic [21], [22], [23], string stability analysis [24], [25]. However, most existing results assume ideal V2V communication which is often not practical. Among many other factors, signal noise in communication channels is a key concern, which can be caused by a variety of internal and/or external factors including limited communication bandwidth [26], cyberattacks [27], etc. If key control parameters, such as time headway, are chosen without consideration of signal noise in communicated signals, then there is a possibility of the onset of string instability and collisions. Although communication quality has improved significantly under 5G and more advanced wireless communication technologies, signal noise when propagating through the platoon could have a substantial effect on safety and performance.

Previous research on imperfect communication focused on how packet drops affected string stability [28]. In [29] and [30], the disturbance estimation and control problem was investigated, though without considering the quantitative effect of disturbance on the design parameters. In contrast to prior research with imperfect communication, this paper considers the following problem: Given a Signal-to-Noise Ratio (SNR) with V2V communication, what is the lower bound of the employable time headway for CACC based vehicle platooning? For this purpose, we consider a CTHP based control law where the feedforward acceleration of the predecessor vehicle is communicated and contains signal noise. Since the communication is through an n -bit channel, we model the noise in the acceleration signal as a sum of n binary random

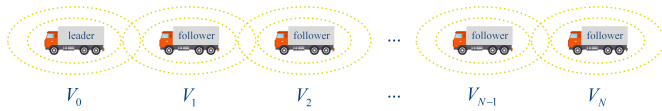


Fig. 1. An illustration of connected and autonomous vehicle platoon with V2V communication.

variables. To address the issue of the vehicle closed-loop governing equation being stochastic due to the noise signal, we develop an equivalent deterministic governing equation using the averaging procedure described in [28]. Based on this, we obtain the inter-vehicular spacing error propagation equation and derive conditions on the control gains and the time headway under which the platoon is robustly string stable in the presence of signal noise and uncertain parasitic actuation lag. We provide numerical simulation results on an illustrative platoon example to verify the main results.

The main contributions of this work can be summarized as follows. In the context of communication from a single predecessor vehicle, this paper provides an extension to our previous work in [9] by considering signal noise in communicated acceleration from the predecessor vehicle. We derive a lower bound on the time headway that is a function of the parasitic actuation lag in the vehicle dynamics (τ_0), acceleration gain (k_a), and SNR (ρ). This lower bound aids in the selection of the time headway in the presence of noise. Further, the derived lower bound of the time headway reduces to that achieved for the ideal communication case given in [9] with no noise in the communicated acceleration signal. In addition, our formulation and analysis allow for a systematic selection of the control gains as well as the time headway for a given signal to noise ratio and an upper bound on the parasitic actuation lag.

The remainder of the paper is organized as follows. Section II contains preliminaries including vehicle dynamics and relevant definitions. The main theoretical results are provided in Section III. An illustrative numerical example and simulation results are provided in Section IV. Finally, some concluding remarks are given in Section V.

II. PRELIMINARIES

Consider a string of autonomous vehicles equipped with V2V communication as illustrated in Fig. 1 where the leader vehicle is indexed by V_0 , the follower vehicles are sequentially indexed by $V_1, V_2, \dots, V_{N-1}, V_N$, where N is the total number of the following vehicles in the platoon. Choosing the moving direction of the platoon as the positive direction, the dynamics of the i -th vehicle is given by the model:

$$\begin{cases} \ddot{x}_i(t) = a_i(t), \\ \tau \dot{a}_i(t) + a_i(t) = u_i(t), \end{cases} \quad (1)$$

where $x_i(t)$, $a_i(t)$, $u_i(t)$ represent the position, acceleration, and control input of the i -th vehicle at time instant t , and $i \in \mathcal{N} = \{1, 2, \dots, N\}$, τ denotes the parasitic actuation lag. It is assumed that τ is *uncertain* with $\tau \in (0, \tau_0]$, where τ_0 is a positive real constant.

Remark 1: Heterogeneity in vehicles can be considered in the vehicle dynamics models which can be feedback linearized with a lower-level controller or compensated through feed-forward into the double integrator model, see [31] and [32].

One could make the argument to consider heterogeneity in the parasitic lag, that is, to consider different parasitic lag for each vehicle, i.e. τ_i for vehicle i . The homogeneous parasitic lag upper bound (τ_0) could be construed as the maximum of all parasitic lags encountered in the vehicles of the platoon, that is $\tau_0 = \max_{i \in \{1, 2, \dots, N\}} \{\tau_i\}$, where N denotes the number of the following vehicles contained in the platoon. It also provides a simpler form of the time headway in terms of the upper bound (τ_0) of the parasitic actuation lag, which makes it easier for the practitioners to select the time headway. This does not mean that all τ_i 's change simultaneously. Such a simplification is often employed in the literature, please see, e.g., [17].

Definition 1: Let d denote the minimum or standstill spacing between adjacent vehicles, $v_i(t)$ denote the velocity of the i -th vehicle, h_w denote the time headway, and $e_i(t) := x_i(t) - x_{i-1}(t) + d$ be the inter-vehicular spacing error of the CSP type for the i -th vehicle with respect to the $(i-1)$ -th vehicle. Define the velocity dependent inter-vehicular spacing error of the CTHP type for the i -th vehicle as:

$$\delta_i(t) = e_i(t) + h_w v_i(t). \quad (2)$$

Definition 2 [9]: Let $\hat{\delta}_i(s)$ denote the Laplace transform of $\delta_i(t)$ and let $H(s)$ denote the inter-vehicular spacing error propagation transfer function that satisfies $\hat{\delta}_i(s) = H(s)\hat{\delta}_{i-1}(s)$. The connected and autonomous vehicle platoon with vehicle dynamics model (1) is said to be robustly string stable if the following two conditions hold for all $\tau \in (0, \tau_0]$: (i) $H(s)$ achieves internal stability and (ii) the platoon is string stable, i.e., it holds that $\|\delta_i(t)\|_\infty \leq \|\delta_{i-1}(t)\|_\infty$, or in the frequency domain

$$\|H(j\omega)\|_\infty \leq 1. \quad (3)$$

Definition 3: Let $s(t)$ and $n(t)$ denote the transmitted signal and the noise signal, respectively, and assume $n(t) = 0$ when $s(t) = 0$. Then, the Signal-to-Noise Ratio (SNR) for $s(t) \neq 0$ is defined as

$$\text{SNR} = \min_{t>0} 20 \log \frac{|s(t)|}{|n(t)|}, \quad (4)$$

where $\log$ denotes the logarithmic function with base 10. From the SNR definition (4), we can obtain that if $\text{SNR} := \rho$, then $\rho = \min_{t>0} \frac{|s(t)|}{|n(t)|} = 10^{\frac{\rho}{20}}$, and for notational convenience, we will use $\rho = \min_{t>0} \frac{|s(t)|}{|n(t)|}$ instead of SNR.

III. MAIN RESULTS

A. Model of Noise

Let $a_{i-1}(t)$ denote the desired or ideal acceleration signal of the $(i-1)$ -th vehicle and $w_{i,i-1}(t) > 0$ denote the noise factor associated with V2V communication of the acceleration signal of the $(i-1)$ -th vehicle to the i -th vehicle. Considering the noise affected signal, the actual communicated acceleration signal available to the i -th vehicle is given by $w_{i,i-1}(t)a_{i-1}(t)$. We define $\rho > 1$ as the signal to noise ratio factor such that $1 - \frac{1}{\rho} \leq w_{i,i-1}(t) \leq 1 + \frac{1}{\rho}$, that is, the communicated acceleration signal to the i -th vehicle satisfies

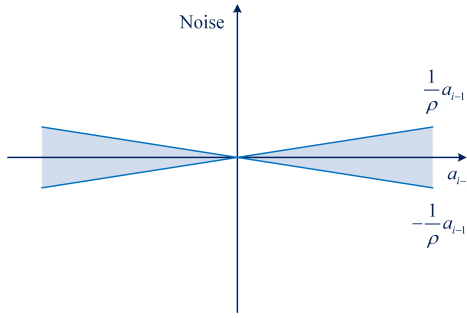


Fig. 2. An illustration of the admissible region for the signal noise in the communicated acceleration signal ($\rho = 5$).

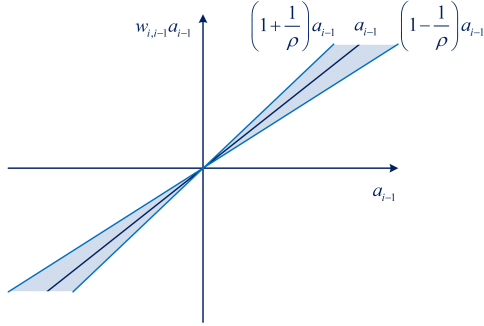


Fig. 3. An illustration of the admissible region of the communicated acceleration signal with respect to the ideal one ($\rho = 5$).

$\left| \left(1 - \frac{1}{\rho}\right) a_{i-1}(t) \right| \leq |w_{i,i-1}(t) a_{i-1}(t)| \leq \left| \left(1 + \frac{1}{\rho}\right) a_{i-1}(t) \right|$. An illustration of the admissible region of the noise signal in terms of the ideal communicated acceleration signal is provided in Fig. 2, and the range of the communicated signal with respect to the ideal signal is provided in Fig. 3.

Considering the communicated acceleration signal is through an n -bit communication channel, we model $w_{i,i-1}(t)$ as follows:

$$w_{i,i-1}(t) = \left(1 - \frac{1}{\rho}\right) + \frac{1}{\rho} \left(\sum_{j=0}^{n-1} \frac{z_{i,j}(t)}{2^j} \right), \quad (5)$$

where $z_{i,j}(t)$, $j = 0, \dots, n-1$, are independent binary random processes. Let $\gamma_{i,j} := \mathbb{E}[z_{i,j}(t)]$ where $\mathbb{E}[\cdot]$ denotes expected value of the argument. We assume that $\gamma_{i,j}$ is independent of time, i.e., the characteristics of the noise processes of the communication channel are time-invariant. However, in order to model a wide variety of noise processes, we do not assume that $\gamma_{i,j}$'s are known a priori. Note that $\gamma_{i,j} \in (0, 1)$ then this noise modeling procedure will allow us to get a communicated signal that takes any value lying between $\left(1 - \frac{1}{\rho}\right) a_{i-1}$ and $\left(1 + \frac{1}{\rho}\right) a_{i-1}$.

B. Inter-Vehicular Spacing Error Propagation Equation

Consider the following CTHP control law for the i -th vehicle:

$$u_i(t) = k_a w_{i,i-1}(t) a_{i-1}(t) - k_v(v_i(t) - v_{i-1}(t)) - k_p \delta_i(t), \quad (6)$$

where k_a , k_v , k_p are positive control gains to be designed. Substituting (6) into (1), then the governing equation for the i -th vehicle is given by:

$$\tau \ddot{x}_i(t) + \ddot{x}_i(t) = k_a w_{i,i-1}(t) a_{i-1}(t) - k_v(v_i(t) - v_{i-1}(t)) - k_p \delta_i(t). \quad (7)$$

The governing equation is stochastic due to the presence of the stochastic noise signal $w_{i,i-1}(t)$; correspondingly, the state is also stochastic. To perform robust string stability analysis, we consider an equivalent deterministic governing equation along the lines described in [28]. By defining the augmented state vector as $\hat{X}(t) = [x_0(t), v_0(t), a_0(t), x_1(t), v_1(t), a_1(t), \dots, x_N(t), v_N(t), a_N(t)]$ and letting $\hat{w}(t)$ be the vector of noise signals with its i -th component being $w_{i,i-1}(t)$, and the acceleration input to the leader vehicle to be $U(t)$, the augmented state equation for the vehicle platoon can be developed as

$$\dot{\hat{X}}(t) = \hat{A}(\hat{w}(t)) \hat{X}(t) + B U(t), \quad (8)$$

where the structures of the matrices $\hat{A}$ and B are given in [28]. Through a discretization process for a sufficiently small time interval, it was shown that

$$\mathbb{E}[\hat{A}] = \bar{A}, \quad \mathbb{E}[e^{\hat{A}t}] = e^{\bar{A}t}, \quad (9)$$

where $\bar{A} = \hat{A}(\bar{w})$ with $\bar{w}$ as the expectation of $\hat{w}(t)$. The second equality is true because of the specific lower triangular and banded structure of the governing equations for the CACC case. Let $\bar{X} = \mathbb{E}[\hat{X}]$; in the following, unless specified otherwise, we will use bar to indicate expected value for a corresponding random variable. The evolution of the vehicle states in the platoon converges to the states of the following deterministic state equation:

$$\dot{\bar{X}}(t) = \bar{A} \bar{X}(t) + B U(t). \quad (10)$$

Utilizing this procedure, the governing equation for the i -th vehicle is given by:

$$\tau \ddot{\bar{x}}_i(t) + \ddot{\bar{x}}_i(t) = k_a \mathbb{E}[w_{i,i-1}(t)] \bar{a}_{i-1}(t) - k_v(\bar{v}_i(t) - \bar{v}_{i-1}(t)) - k_p \bar{\delta}_i(t). \quad (11)$$

Define $\tilde{k}_a$ as

$$\tilde{k}_a := k_a \mathbb{E}[w_{i,i-1}(t)] = \left[1 - \frac{1}{\rho} + \frac{1}{\rho} \left(\sum_{j=0}^{n-1} \frac{\gamma_{i,j}}{2^j} \right) \right] k_a, \quad (12)$$

then the governing equation for the i -th vehicle given by (11) can be rewritten as follows:

$$\tau \ddot{\bar{x}}_i(t) + \ddot{\bar{x}}_i(t) = \tilde{k}_a \ddot{\bar{x}}_{i-1}(t) - k_v(\ddot{\bar{x}}_i(t) - \ddot{\bar{x}}_{i-1}(t)) - k_p \bar{\delta}_i(t). \quad (13)$$

Let $\bar{\delta}_i(t) = \bar{x}_i(t) - \bar{x}_{i-1}(t) + d + h_w \dot{\bar{x}}_i(t)$, and let $\bar{\delta}_i(s)$ be the Laplace transform of $\bar{\delta}_i(t)$. Then, the inter-vehicular spacing error propagation equation can be derived as

$$\bar{\delta}_i(s) = \tilde{H}(s; \tau) \bar{\delta}_{i-1}(s), \quad (14)$$

where $\tilde{H}(s; \tau) = \tilde{N}(s)/\mathcal{D}(s)$ is the inter-vehicular spacing error propagation transfer function with

$$\begin{aligned}\tilde{N}(s) &= \tilde{k}_a s^2 + k_v s + k_p, \\ \mathcal{D}(s) &= \tau s^3 + s^2 + \gamma s + k_p,\end{aligned}$$

where $\gamma := k_v + h_w k_p$.

The platoon error propagation equation given by (13) was studied in Theorem 1 of [9] which can be recast for the case considered in this paper as follows:

Theorem 1: The following are true for the platoon given by the inter-vehicular spacing error propagation equation (14):

- (a). $\|\tilde{H}(j\omega; \tau)\|_\infty \leq 1, \forall \tau \in (0, \tau_0]$, implies that $\tilde{k}_a \in (0, 1)$;
- (b). given any $\tilde{k}_a \in (0, 1)$, to ensure there exist $k_v, k_p > 0$ such that $\|\tilde{H}(j\omega; \tau)\|_\infty \leq 1$ for all $\tau \in (0, \tau_0]$, h_w must satisfy the following:

$$h_w > \frac{2\tau_0}{1 + \tilde{k}_a}. \quad (15)$$

C. Stability Analysis

Theorem 2: The following are true for the platoon given by the inter-vehicular spacing error propagation equation (14):

- (a). $\|\tilde{H}(j\omega; \tau)\|_\infty \leq 1, \forall \tau \in (0, \tau_0]$, implies that $\tilde{k}_a \in (0, 1)$;
- (b). given any

$$k_a \in \left(0, \frac{1}{1 + \frac{1}{\rho}}\right), \quad (16)$$

and h_w satisfying the following:

$$h_w > h_{w,lb}(k_a) := 2\tau_0 \left(\frac{1 - \left(1 - \frac{1}{\rho}\right)k_a}{1 - \left(1 + \frac{1}{\rho}\right)^2 k_a^2} \right), \quad (17)$$

there exist $k_v, k_p > 0$ such that $\|\tilde{H}(j\omega; \tau)\|_\infty \leq 1$ for all $\tau \in (0, \tau_0]$;

- (c). given any $\rho > 1$, the minimizing value, k_a^* , of k_a and the corresponding minimum value, $h_{w,lb}^*$, of $h_{w,lb}(k_a)$ are respectively given by:

$$k_a^* = \left(\frac{1 - \frac{1}{\sqrt{\rho}}}{1 + \frac{1}{\sqrt{\rho}}} \right) \frac{1}{\left(1 + \frac{1}{\rho}\right)}, \quad (18)$$

$$h_{w,lb}^* = \tau_0 \frac{\left(1 + \frac{1}{\sqrt{\rho}}\right)^2}{\left(1 + \frac{1}{\rho}\right)}. \quad (19)$$

Correspondingly, there exist $k_v, k_p > 0$ such that $\|\tilde{H}(j\omega; \tau)\|_\infty \leq 1$ for all $\tau \in (0, \tau_0]$.

Remark 2: Minimizing $h_{w,lb}$ is useful for improving traffic throughput because a lower time headway can be employed while still guaranteeing “robust” string stability.

Proof: For internal stability of (14), employing Routh-Hurwitz criterion on $\mathcal{D}(s)$ yields the internal stability condition as $\gamma - \tau k_p > 0$. Then (14) is robustly internally stable for all $\tau \in (0, \tau_0]$ iff $\gamma - \tau_0 k_p > 0$. For string stability, applying Theorem 1 to the inter-vehicular spacing error propagation equation (14), we infer that:

- (a). $\tilde{k}_a \in (0, 1)$;
- (b). $h_w \geq \frac{2\tau_0}{1 + \tilde{k}_a}$.

Since $\gamma_{i,j}$'s are not known, we can bound $\tilde{k}_a$ by considering their maximum possible values. Hence, we have

$$\tilde{k}_a \leq \left(1 - \frac{1}{\rho} + \frac{2}{\rho}\right)k_a = \left(1 + \frac{1}{\rho}\right)k_a. \quad (20)$$

Thus, if

$$k_a < \frac{1}{1 + \frac{1}{\rho}} \implies \tilde{k}_a < 1, \quad (21)$$

allows us to apply part (b) of Theorem 1, we then have

$$h_w \geq \frac{2\tau_0}{1 + \tilde{k}_a}. \quad (22)$$

Again, since $\tilde{k}_a$ is dependent on $\gamma_{i,j}$'s which are not known a priori, we will consider the worst possible lower bound, i.e., the lower bound for h_w that is a maximum over all possible values of $\gamma_{i,j}$'s. To address this, we require

$$h_w > \max_{\gamma_{i,0}, \dots, \gamma_{i,n-1}} \frac{2\tau_0}{1 + \tilde{k}_a}. \quad (23)$$

Hence, if the above condition holds, then (22) will hold. Substituting $\tilde{k}_a$ from (12) into (23), the condition (22) will hold if

$$\begin{aligned}h_w &> \max_{\gamma_{i,0}, \dots, \gamma_{i,n-1}} \frac{2\tau_0}{1 + \left(1 - \frac{1}{\rho} + \frac{1}{\rho} \left(\sum_{j=0}^{n-1} \frac{\gamma_{i,j}}{2^j} \right) \right) k_a} \\ &= \frac{2\tau_0}{\min_{\gamma_{i,0}, \dots, \gamma_{i,n-1}} \left(1 + \left(1 - \frac{1}{\rho} + \frac{1}{\rho} \left(\sum_{j=0}^{n-1} \frac{\gamma_{i,j}}{2^j} \right) \right) k_a \right)} \\ &= \frac{2\tau_0}{1 + \left(1 - \frac{1}{\rho}\right)k_a}. \quad (24)\end{aligned}$$

In [9], the feasible region for the control gains k_v and k_p is specific to one chosen value of $\tilde{k}_a \in (0, 1)$. However, in this case, $\tilde{k}_a$ could be any value in the interval

$$\mathcal{I} := \left[\left(1 - \frac{1}{\rho}\right)k_a, \left(1 + \frac{1}{\rho}\right)k_a \right]$$

for a given $k_a \in \left(0, \frac{1}{1 + \frac{1}{\rho}}\right)$. Let $\mathcal{F}(\tilde{k}_a)$ denote the set of all (k_v, k_p) that ensures robust string stability for a given value of $\tilde{k}_a$. Unlike in [9], we need to show that the set of all feasible (k_v, k_p) that works for any $\tilde{k}_a \in \mathcal{I}$ denoted by $\mathcal{S}$ is non-empty, i.e.,

$$\mathcal{S} := \bigcap_{\tilde{k}_a \in \mathcal{I}} \mathcal{F}(\tilde{k}_a) \neq \emptyset.$$

In the remainder of the proof, we will focus on showing that $\mathcal{S} \neq \emptyset$ and explicitly construct this set for synthesizing the controller. We do so by considering the robust string stability condition, $\|\tilde{H}(j\omega; \tau)\|_\infty \leq 1, \forall \tau \in (0, \tau_0]$ and employing a time headway satisfying (24).

According to *Definition 2*, robust string stability is guaranteed when $\|\tilde{H}(j\omega; \tau)\|_\infty^2 \leq 1$, i.e., $|\tilde{N}(j\omega)|^2 \leq |\mathcal{D}(j\omega)|^2$, $\forall \omega$, which can be rewritten as

$$\tau^2 \omega^4 + (1 - \tilde{k}_a^2 - 2\tau\gamma)\omega^2 + \gamma^2 - 2k_p - k_v^2 + 2\tilde{k}_a k_p \geq 0. \quad (25)$$

The above condition is satisfied if $\forall \tau \in (0, \tau_0]$, it holds that

$$\begin{cases} 1 - \tilde{k}_a^2 - 2\tau\gamma \geq 0, \\ \gamma^2 - 2k_p - k_v^2 + 2\tilde{k}_a k_p \geq 0, \end{cases} \quad (26a)$$

i.e.,

$$\begin{cases} \gamma \leq \frac{1 - \tilde{k}_a^2}{2\tau_0}, \\ \gamma \geq \sqrt{2k_p(1 - \tilde{k}_a) + k_v^2}. \end{cases} \quad (27a)$$

$$\gamma \geq \sqrt{2k_p(1 - \tilde{k}_a) + k_v^2}. \quad (27b)$$

Thus, by considering the appropriate lower and upper bounds of $\tilde{k}_a$ in the inequalities (27a) and (27b), these inequalities on γ can be satisfied for all $\tilde{k}_a \in \mathcal{I}$ if γ satisfies

$$\begin{cases} \gamma \leq \min_{\tilde{k}_a \in \mathcal{I}} \frac{1 - \tilde{k}_a^2}{2\tau_0} = \frac{1 - \left(1 + \frac{1}{\rho}\right)^2 k_a^2}{2\tau_0}, \\ \gamma \geq \max_{\tilde{k}_a \in \mathcal{I}} \sqrt{2k_p(1 - \tilde{k}_a) + k_v^2} \end{cases} \quad (28a)$$

$$= \sqrt{2k_p \left(1 - \left(1 - \frac{1}{\rho}\right)k_a\right) + k_v^2}, \quad (28b)$$

from which the admissible range of γ is given by

$$\sqrt{2k_p \left(1 - \left(1 - \frac{1}{\rho}\right)k_a\right) + k_v^2} \leq \gamma \leq \frac{1 - \left(1 + \frac{1}{\rho}\right)^2 k_a^2}{2\tau_0}. \quad (29)$$

Next, we will show that the set $\mathcal{S} \neq \emptyset$. First, since $\gamma = k_v + h_w k_p$, the right inequality in (29) can be rewritten as

$$k_v + h_w k_p \leq \frac{1 - \left(1 + \frac{1}{\rho}\right)^2 k_a^2}{2\tau_0}. \quad (30)$$

Thus, the admissible set of k_v and k_p for (30) is given by

$$\mathcal{S}_1 := \left\{ (k_v, k_p) : k_v > 0, k_p > 0, \frac{k_v}{a_1} + \frac{k_p}{b_1} \leq 1 \right\}, \quad (31)$$

where a_1 and b_1 are respectively defined as

$$\begin{cases} a_1 = \frac{1 - \left(1 + \frac{1}{\rho}\right)^2 k_a^2}{2\tau_0}, \\ b_1 = \frac{1 - \left(1 + \frac{1}{\rho}\right)^2 k_a^2}{2\tau_0 h_w}. \end{cases}$$

Second, the left inequality in (29) can be rewritten as

$$k_v + h_w k_p \geq \sqrt{2k_p \left(1 - \left(1 - \frac{1}{\rho}\right)k_a\right) + k_v^2}. \quad (32)$$

Squaring both sides of (32) and simplifying, we obtain

$$2h_w k_v + h_w^2 k_p \geq 2 \left(1 - \left(1 - \frac{1}{\rho}\right)k_a\right). \quad (33)$$

Thus, the admissible set of k_v and k_p for (32) is given by

$$\mathcal{S}_2 := \left\{ (k_v, k_p) : k_v > 0, k_p > 0, \frac{k_v}{a_2} + \frac{k_p}{b_2} \geq 1 \right\}, \quad (34)$$

where a_2 and b_2 are respectively defined as

$$\begin{cases} a_2 = \frac{1 - \left(1 - \frac{1}{\rho}\right)k_a}{h_w}, \\ b_2 = \frac{2 \left(1 - \left(1 - \frac{1}{\rho}\right)k_a\right)}{h_w^2}. \end{cases}$$

Then, combining (31) and (34), the feasible region of k_v and k_p for (29) is given by the set

$$\mathcal{S} = \mathcal{S}_1 \cap \mathcal{S}_2. \quad (35)$$

Note that $\mathcal{S}$ is nonempty if $a_1 \geq a_2$ or $b_1 \geq b_2$. In particular, notice that

$$\frac{a_1}{a_2} = \frac{h_w}{2\tau_0} \left(\frac{1 - \left(1 + \frac{1}{\rho}\right)^2 k_a^2}{1 - \left(1 - \frac{1}{\rho}\right)k_a} \right).$$

Substituting h_w from (17), we have

$$\frac{a_1}{a_2} > 1.$$

Thus, $\mathcal{S} \neq \emptyset$. This completes the proof of Statement (b) of *Theorem 2*. In the following, we prove Statement (c) of *Theorem 2*.

Let $\bar{h}_w(k_a) := \frac{h_w, lb(k_a)}{2\tau_0}$, $m := 1 - \frac{1}{\rho}$, and $n := \left(1 + \frac{1}{\rho}\right)^2$. Then,

$$\bar{h}_w(k_a) = \frac{1 - m k_a}{1 - n k_a^2} \quad (36)$$

and

$$\frac{d\bar{h}_w(k_a)}{dk_a} = \frac{-m n k_a^2 + 2 n k_a - m}{(1 - n k_a^2)^2}. \quad (37)$$

Let $f(k_a) := -m n k_a^2 + 2 n k_a - m$. Then, the roots of $f(k_a) = 0$ are given by

$$r_{1,2} := \frac{n \mp \sqrt{n(n - m^2)}}{m n}.$$

Note that $r_1 < \frac{1}{1 + \frac{1}{\rho}}$ and $r_2 > \frac{1}{1 + \frac{1}{\rho}}$. The function $f(k_a)$ vs. k_a is provided in Fig. 4 as an illustration for $\rho = 5$. Since $k_a \in \left(0, \frac{1}{1 + \frac{1}{\rho}}\right)$, we need to consider only r_1 . Hence, we have

$$\begin{cases} \frac{d\bar{h}_w}{dk_a} < 0, \text{ when } k_a \in (0, r_1); \\ \frac{d\bar{h}_w}{dk_a} > 0, \text{ when } k_a \in \left(r_1, \frac{1}{1 + \frac{1}{\rho}}\right). \end{cases} \quad (38)$$

Thus,

$$\min_{k_a \in \left(0, \frac{1}{1 + \frac{1}{\rho}}\right)} \bar{h}_w(k_a) = \bar{h}_w(r_1). \quad (39)$$

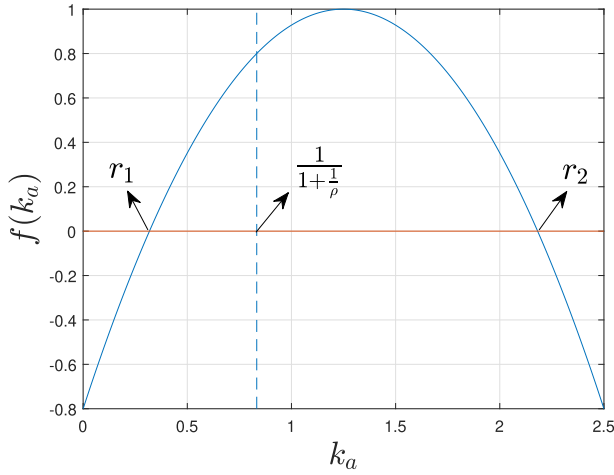


Fig. 4. $f(k_a)$ vs. k_a when $\rho = 5$.

Substituting m and n into r_1 , we obtain

$$k_a^* := r_1 = \left(\frac{1 - \frac{1}{\sqrt{\rho}}}{1 + \frac{1}{\sqrt{\rho}}} \right) \left(\frac{1}{1 + \frac{1}{\rho}} \right). \quad (40)$$

Further, substituting r_1 into $\bar{h}_w(r_1)$, we obtain

$$\bar{h}_w(r_1) = \frac{\left(1 + \frac{1}{\sqrt{\rho}}\right)^2}{2\left(1 + \frac{1}{\rho}\right)}. \quad (41)$$

Therefore,

$$h_{w,lb}^* = 2\tau_0 \bar{h}_w(r_1) = \tau_0 \frac{\left(1 + \frac{1}{\sqrt{\rho}}\right)^2}{1 + \frac{1}{\rho}}. \quad (42)$$

This completes the proof of Statement (c) of *Theorem 2*.

Finally, we will prove the lower bound of the time headway given by (19) satisfies the internal stability condition. For this purpose, note that

$$h_w k_p \geq \tau_0 k_p \frac{\left(1 + \frac{1}{\sqrt{\rho}}\right)^2}{\left(1 + \frac{1}{\rho}\right)}. \quad (43)$$

Thus, $\gamma = k_v + h_w k_p > h_w k_p \geq \tau_0 k_p$.

Therefore, this completes the proof of *Theorem 2*. ■

Remark 3: If $\rho \rightarrow \infty$ (no signal noise case), then $k_a \in (0, 1)$ and $h_{w,lb}(k_a) = \frac{2\tau_0}{1+k_a}$ as given in [9].

Remark 4: In (17), if $\rho \rightarrow \infty$, then the lower bound of the time headway reduces to $h_w > h_{w,lb}(k_a) = \frac{2\tau_0}{1+k_a}$ as given in [9]. For the same k_a value, the lower bound of the time headway increases as ρ decreases in the presence of signal noise. In addition, according to (16), the upper bound of k_a is smaller when compared to the no noise case, which also factors into the increase of the lower bound of the time headway in the presence of noise. In addition, in (19), if $\rho \rightarrow \infty$, the minimum lower bound of the time headway reduces to $h_{w,lb}^* \rightarrow \tau_0$, and the corresponding optimal k_a given by (18) becomes $k_a^* \rightarrow 1$.

TABLE I

THE VALUES OF $\gamma_{i,j}$, $j = 0, 1, 2, \dots, 15$

$\gamma_{i,0}$	$\gamma_{i,1}$	$\gamma_{i,2}$	$\gamma_{i,3}$	$\gamma_{i,4}$	$\gamma_{i,5}$
0.8055	0.5767	0.1829	0.2399	0.8865	0.0287
$\gamma_{i,6}$	$\gamma_{i,7}$	$\gamma_{i,8}$	$\gamma_{i,9}$	$\gamma_{i,10}$	$\gamma_{i,11}$
0.4899	0.1679	0.9787	0.7127	0.5005	0.4711
$\gamma_{i,12}$	$\gamma_{i,13}$	$\gamma_{i,14}$	$\gamma_{i,15}$		
0.0596	0.6820	0.0424	0.0714		

Remark 5: Note that the deterministic system (13) is an equivalent system to the stochastic system (11) in the mean or expected sense, and robust string stability analysis is performed on the deterministic system (13).

IV. NUMERICAL SIMULATIONS

In this section, we consider a numerical example of a platoon and provide simulation results to corroborate the main theoretical results achieved in Section III. We first consider numerical values of essential parameters and an acceleration maneuver profile for the lead vehicle that are meaningful to evaluate the results. Then, we calculate numerical values for the upper bound of the acceleration gain (k_a) and the lower bound of the time headway (h_w). We select the acceleration gain and time headway values to satisfy these bounds. Lastly, we provide and discuss the simulation results for several cases by selecting acceleration gain and time headway values satisfying the bounds, and position and velocity gains lying in the feasible sets.

A. Numerical Values of Parameters and Lead Vehicle Acceleration Maneuver Profile

We consider a platoon consisting of a lead vehicle and 12 following vehicles, that is, $N = 12$ with the following numerical values for the parasitic actuation lag upper bound (τ_0) and standstill spacing distance (d): $\tau_0 = 0.5$ s and $d = 5$ m. The steady-state velocity of the platoon is considered as 25 m/s (90 km/hr). In the vehicle models for the numerical simulations, we choose the numerical value for τ as τ_0 . We assume that the lead vehicle experiences an external disturbance which causes a perturbation on its acceleration, denoted as $a_0(t)$, as follows:

$$a_0(t) = \begin{cases} 0.5 \sin(0.1(t - 10)), & 10 \text{ s} < t < (10 + 20\pi) \text{ s}, \\ 0, & \text{otherwise.} \end{cases}$$

From the above, note that this acceleration maneuver is initiated at the 10 s mark of the simulation.

For the V2V communicated acceleration from the predecessor vehicle signal, we consider 16 communication channels or discretizations, that is, $n = 16$ and the signal to noise ratio factor of t , that is, $\rho = 5$. The expectations ($\gamma_{i,j}$) of the 16 binary random variables $z_{i,j}$ are provided in TABLE I.

Then, according to (16) in *Theorem 2*, the upper bound of k_a can be computed as

$$k_a < \frac{1}{1 + \frac{1}{\rho}} = 0.8333. \quad (44)$$

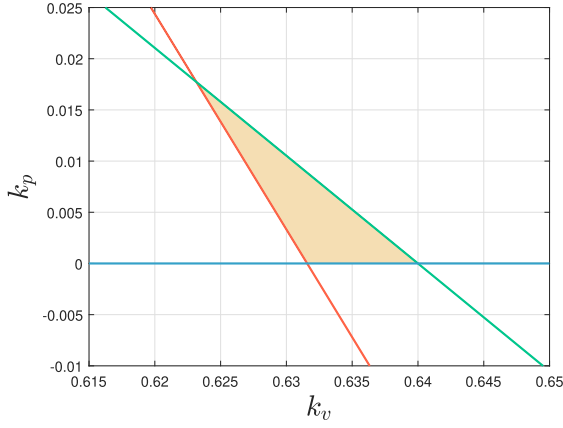


Fig. 5. Case 1: The feasible region of k_v and k_p when $k_a = 0.5$ and $h_w = 0.95$ s.

Choosing $k_a = 0.5$ and substituting τ_0, ρ, k_a values in (17), we can derive the time headway lower bound as

$$h_{w,lb} = 2\tau_0 \left(\frac{1 - \left(1 - \frac{1}{\rho}\right)k_a}{1 + \left(1 + \frac{1}{\rho}\right)^2 k_a^2} \right) = 0.9375 \text{ s.} \quad (45)$$

B. Simulation Results

We will consider five cases for the numerical simulations:

- 1) **Case 1:** $h_w = 0.95$ s; this satisfies the time headway lower bound constraint.
- 2) **Case 2:** $h_w = 0.65$ s; this violates the time headway lower bound constraint, i.e., this value is smaller than the lower bound.
- 3) **Case 3:** $k_a = k_a^* = 0.3183$ and $h_w = h_{w,lb}^*$; these values correspond to the minimum values of k_a and h_w given by Eqs. (18) and (19) in Statement (c) of Theorem 3.2.
- 4) **Case 4:** Evaluation of total length of the platoon for two different time headway values, $h_w = 0.88$ s and $h_w = 0.95$ s.
- 5) **Case 5:** Evaluation of heterogeneity in vehicle models by considering different parasitic lags for platoon vehicles ($h_w = 0.95$ s).

Case 1: Choosing $h_w = 0.95$ s, then based on (35), the feasible region of k_v and k_p is obtained as shown in Fig. 5. From this region, we choose $k_v = 0.63$, $k_p = 0.009$ for the numerical simulations. With the above chosen values of the time headway and control gains, when $\tilde{k}_a$ varies in the interval $[(1 - \frac{1}{\rho})k_a, (1 + \frac{1}{\rho})k_a]$, the frequency response of $|\tilde{H}(j\omega; \tau_0)|$ is shown in Fig. 6, which demonstrates string stability of the platoon since $\max |\tilde{H}(j\omega; \tau_0)| \leq 1$, $\omega \geq 0$.

The evolution of the inter-vehicular spacing errors is shown in Fig. 7. The platoon spacing errors are bounded and converge to zero in response to the acceleration maneuver of the lead vehicle. Further, the maximum spacing error between vehicles decreases as we traverse from the lead vehicle down to the last vehicle in the platoon. In addition, as an example, under the noise model (5) with the expectations of the binary random variables $z_{i,j}$ as provided in TABLE I, the noise and the

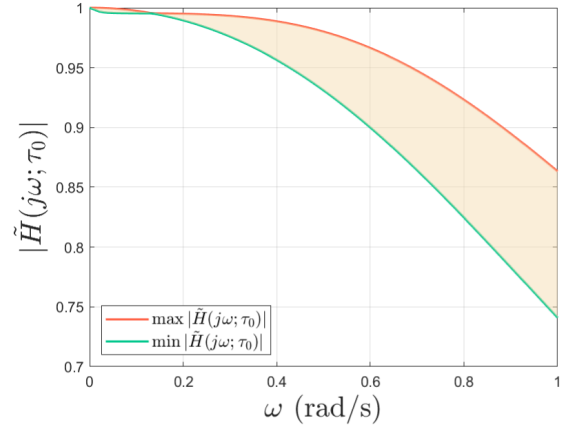


Fig. 6. Case 1: $|\tilde{H}(j\omega; \tau_0)|$ when $k_a = 0.5$, $h_w = 0.95$ s, $k_v = 0.63$, and $k_p = 0.009$.

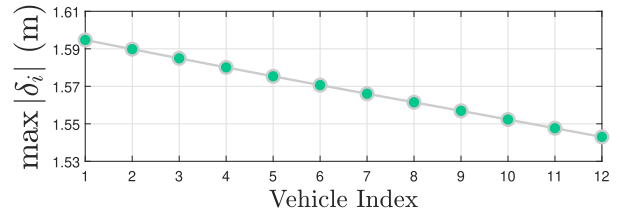
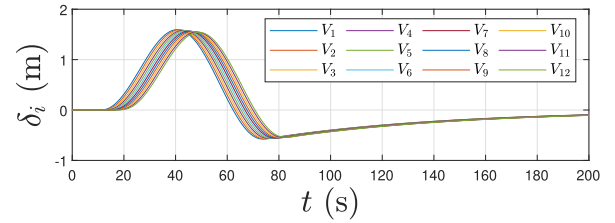


Fig. 7. Case 1: The inter-vehicular spacing errors (top) and the maximum of the spacing errors of the following vehicles for the given lead vehicle acceleration maneuver ($k_a = 0.5$, $h_w = 0.95$ s, $k_v = 0.63$, $k_p = 0.009$).

communicated acceleration signal from vehicle 1 to vehicle 2 are shown in Fig. 8.

Case 2: In Case 1, we considered the time headway and all gains satisfying the main results in Section III. For comparison, to evaluate the tightness of the lower bound, in Case 2, we consider the same gains but a value of the time headway smaller than the lower bound given in Section III for robust string stability. We choose $h_w = 0.65$ s, which is smaller than the time headway lower bound of $h_{w,lb} = 0.9375$ s for $k_a = 0.5$. For this case, when $\tilde{k}_a$ varies in the interval $[(1 - \frac{1}{\rho})k_a, (1 + \frac{1}{\rho})k_a]$, the frequency response of $|\tilde{H}(j\omega; \tau_0)|$ is shown in Fig. 9 which exhibits string instability since $\max |\tilde{H}(j\omega; \tau_0)| > 1$, $\omega \geq 0$; note that for lower frequencies, the maximum amplitude of the spacing error transfer function is greater than unity.

The inter-vehicular spacing errors corresponding to Case 2 are shown in Fig. 10; note that the maximum spacing error increases as we traverse from the lead vehicle to the last vehicle in the platoon.

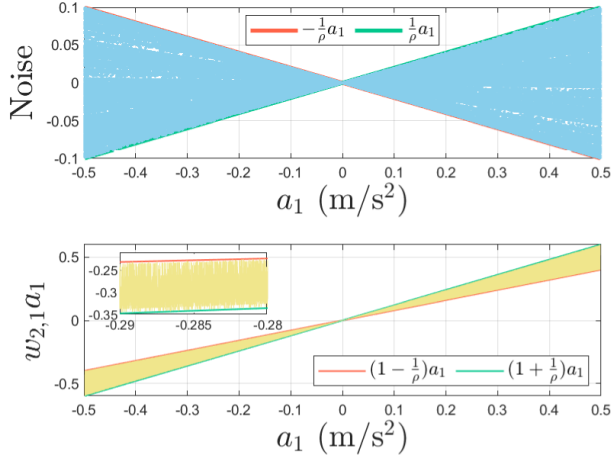


Fig. 8. Case 1: The noise and the communicated acceleration signal from vehicle 1 to vehicle 2 ($k_a = 0.5$, $h_w = 0.95$ s, $k_v = 0.63$, $k_p = 0.009$).

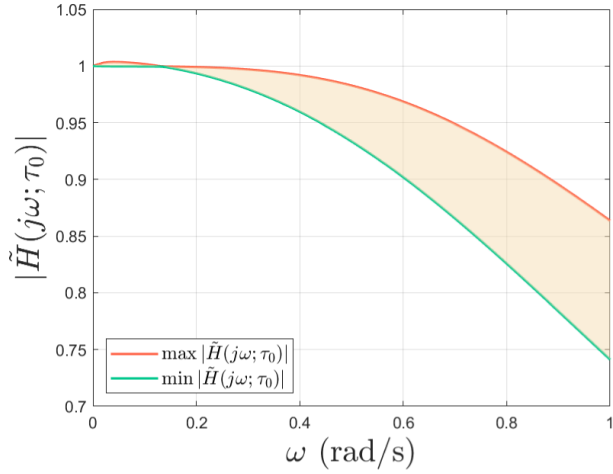


Fig. 9. Case 2: $|\tilde{H}(j\omega; \tau_0)|$ when $k_a = 0.5$, $h_w = 0.65$ s, $k_v = 0.63$, and $k_p = 0.009$.

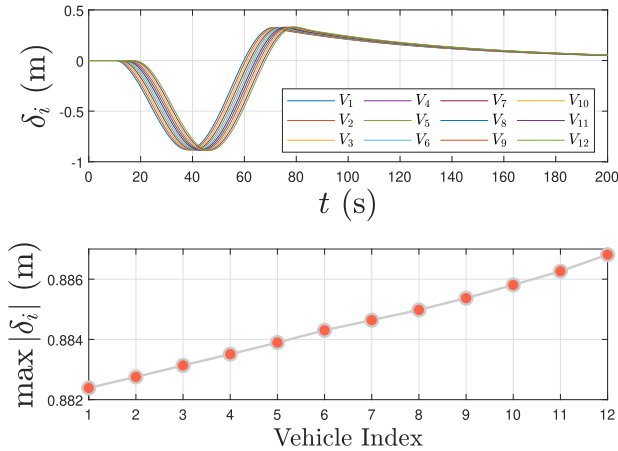


Fig. 10. Case 2: The inter-vehicular spacing errors of the following vehicles ($k_a = 0.5$, $k_v = 0.63$, $k_p = 0.009$, $h_w = 0.65$ s).

Case 3: In this case, numerical simulations were conducted to evaluate the performance of the platoon for k_a^* and $h_{w,lb}^*$ given in Statement (c) of *Theorem 2*. For $\rho = 5$, according

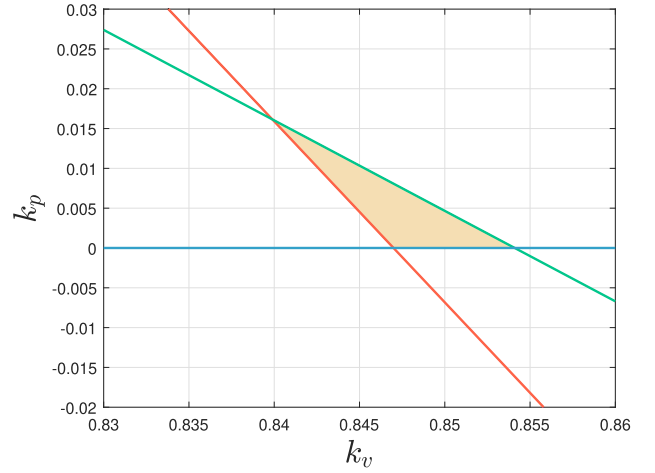


Fig. 11. Case 3: The feasible region of k_v and k_p when $k_a = k_a^*$, $h_w = 0.88$ s.

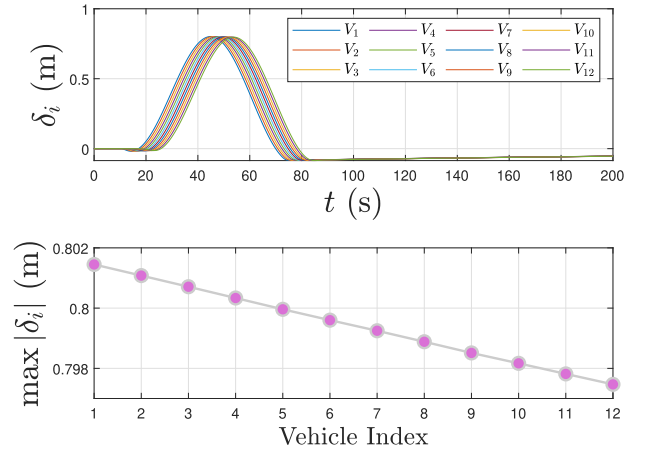


Fig. 12. Case 3: The inter-vehicular spacing errors of the following vehicles ($k_a = k_a^*$, $h_w = 0.88$ s, $k_v = 0.85$, $k_p = 0.003$).

to (18) and (19), we can obtain

$$k_a^* = \left(\frac{1 - \frac{1}{\sqrt{\rho}}}{1 + \frac{1}{\sqrt{\rho}}} \right) \frac{1}{\left(1 + \frac{1}{\rho}\right)} = 0.3183,$$

and

$$h_{w,lb}^* = \tau_0 \frac{\left(1 + \frac{1}{\sqrt{\rho}}\right)^2}{\left(1 + \frac{1}{\rho}\right)} = 0.8727.$$

Choosing $k_a = k_a^*$ and $h_w = 0.88$, the feasible region of k_v and k_p is shown in Fig. 11. Choosing $k_v = 0.85$, $k_p = 0.003$ from the feasible region, the evolution of the inter-vehicular spacing errors is provided in Fig. 12. Note that as expected the maximum spacing errors are smaller than those observed in Case 1.

Case 4: In this case, we numerically evaluate the effect of the time headway selection on the total platoon length. Figure 13 provides the total length of the platoon ($x_0 - x_N$) for two time headway values, $h_w = 0.95$ s and $h_w = 0.88$ s. The length of the platoon is smaller for smaller time headways, indicating a higher throughput for smaller time headways as commented in *Remark 2*.

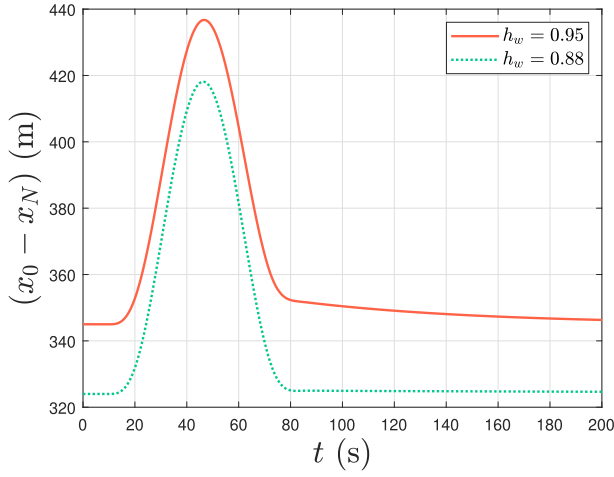


Fig. 13. Case 4: The comparison of the length of the platoon between $h_w = 0.95$ and $h_w = 0.88 \approx h_{w,lb}^*$.

TABLE II
THE VALUES OF $\tau_i, i = 1, 2, \dots, 12$

τ_1	τ_2	τ_3	τ_4	τ_5	τ_6
0.2530	0.1111	0.1535	0.3987	0.1253	0.0562
τ_7	τ_8	τ_9	τ_{10}	τ_{11}	τ_{12}
0.2240	0.4101	0.1998	0.2375	0.0257	0.1761

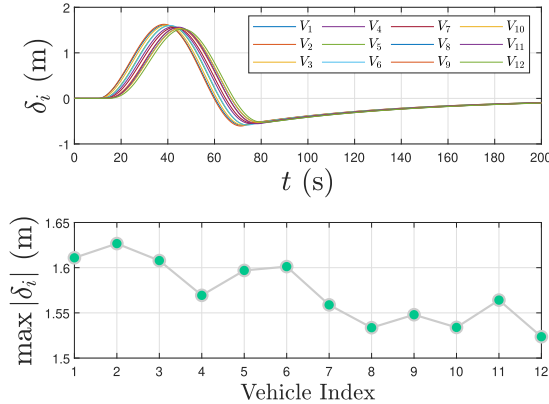


Fig. 14. Case 5 (Heterogeneous parasitic actuation lags as given in TABLE II): The inter-vehicular spacing errors (top) and the maximum of the spacing errors of the following vehicles for the given lead vehicle acceleration maneuver ($k_a = 0.5$, $h_w = 0.95$, $k_v = 0.63$, $k_p = 0.009$).

Case 5: In this case, we numerically evaluate the effectiveness of the selected time headway and control gains on heterogeneous vehicle models by assuming the parasitic actuation lag of vehicle i is denoted by τ_i . We randomly generated values of τ_i in the range of $(0, \tau_0]$, which are provided in TABLE II and conducted simulations with the parameter values in Case 1, and with the same values of the control gains and the time headway, the responses of the inter-vehicular spacing errors are provided in Fig. 14. In addition, the responses of the accelerations of the following vehicles are given in Fig. 15.

From the extensive numerical simulations we have conducted and a representative sample of those presented and discussed here, we have corroborated the main theoretical results, including robust string stability and various remarks.

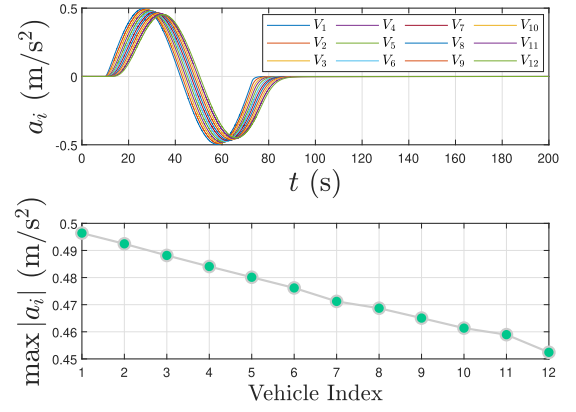


Fig. 15. Case 5 (Heterogeneous parasitic actuation lags as given in TABLE II): The accelerations of the following vehicles (top) and the maximum of the accelerations of the following vehicles for the given lead vehicle acceleration maneuver ($k_a = 0.5$, $h_w = 0.95$, $k_v = 0.63$, $k_p = 0.009$).

V. SUMMARY AND CONCLUSION

We have investigated robust string stability of connected and autonomous vehicle platoons with cooperative adaptive cruise control systems subject to noisy V2V communication. We have derived conditions on control gains for predecessor acceleration, relative velocity and spacing error and a lower bound for the time headway that are dependent on the signal to noise ratio due to V2V communication of acceleration, while ensuring both internal and string stability of the CACC system. We have provided a systematic analysis through which one can select control gains and time headway for a given SNR. The results clearly layout the effect of communication noise in vehicle platoons and provide a clear path to practitioners to select time headway and controller gains in cooperative adaptive cruise control systems in the presence of communication noise. The novelty of the noise model and the systematic approach provide new tools for researchers to address similar noise related problems in connected and autonomous vehicles. We have also provided an illustrative example and corresponding simulation results to demonstrate the main results. For future work, we will investigate extension under noisy communication when multiple predecessor follower information flow is employed. We also plan to discuss the results with practitioners and test the effectiveness of the obtained results in their platoon implementations.

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